A054 v0.5.1: E013 Provider-Convergent State-Selection Falsifier

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2026

1 Identification

Catalog ID A054; release v0.5.1; SST Falsifier Framework v1.0.6, profile `multilibrary_gpu`. This is an E013 cross-falsifier member at stage 30.

2 Question

Can distinct current E011/PKLSA provider representatives of the same topology be mapped by one target-free variational selector to a common stationary and locally stable finite-core state?

3 Hypotheses

The null states that provider dependence survives target-free state selection or that selected stationary states fail local stability. The alternative requires provider-convergent stationary and stable endpoints without topology/provider-specific retuning.

4 Assumptions

E013 fixes the current carrier population. The selector is a variational existence/selection test, not a claim that inviscid Euler flow is dissipative. Physical stability is tested separately after stationarity. Legacy falsifier results are excluded as evidence.

5 Symbols

$\widehat{H}_a$ is the reach-normalized regularized line-energy kernel, q Bishop-frame deformation coordinates, σ the normalized energy-gradient stationarity score, and K_H the endpoint Hessian.

6 Equations

Machine-readable science contract

Research question On one E013-frozen population of current E011 provider anchors, can a fixed target-free variational state-selection procedure map distinct source representatives of the same topology to a common stationary and locally stable finite-core state, without using legacy falsifier results?

Objective Re-run state selection from source-native bytes for exactly the E013 carriers using one canonical constrained objective Uc(q)=H(N[R(X+deltaX(q))]) for gradient, Armijo line search, Hessian and ringdown; quantify stationarity, topology preservation, provider-endpoint convergence and local stability; numerical inability to demonstrate descent or Hessian-sign convergence is UNRESOLVED rather than a scientific FAIL; freeze only provider-converged stable endpoints as a future A057 handoff.

H_0 H0: provider-distinct current PKLSA/SKLSA representatives do not converge under the preregistered selector to a reproducible stationary and stable topology-level state beyond numerical/source variation.

H_1 H1: the same preregistered selector, without topology/provider-specific retuning, yields provider-convergent stationary and locally stable endpoints for a reproducible subset of topologies.

Registered equations

E1: Regularized finite-core comparison Hamiltonian

$$\widehat{H}_a = \sum_{i,j} \oint \oint \frac{d\widehat{\mathbf{X}}_i \cdot d\widehat{\mathbf{X}}_j}{\sqrt{\|\widehat{\mathbf{X}}_i - \widehat{\mathbf{X}}_j\|^2 + \widehat{a}^2}} \tag{1}$$

Dimensional check: All lengths are reach-normalized; the kernel is dimensionless.

E2: Bishop-Fourier reduced deformation

$$\delta\mathbf{X}(q) = \sum_{\mu} q_\mu \mathbf{B}_\mu, \qquad \mathbf{B}_\mu \perp \mathbf{t} \tag{2}$$

Dimensional check: In reach-normalized coordinates both q and B are dimensionless.

E3: Canonical constrained objective and stationarity score

$$\mathcal{P} = \mathcal{N} \circ \mathcal{R}, \qquad U_c(q; X) = \widehat{H}_a(\mathcal{P}[X + \delta X(q)]), \qquad \sigma = \frac{\|\nabla_q U_c(0; X)\|_2}{\max(|U_c(0; X)|, \epsilon)} \tag{3}$$

Dimensional check: Projection and the reach-normalized Hamiltonian are dimensionless, so U_c and sigma are dimensionless.

E4: Target-free Armijo/trust-region selector on the constrained objective

$$d_n = -\frac{\nabla U_c}{\|\nabla U_c\|}, \qquad \eta_{n,k} = \min(\Delta_n, \eta_0)\beta^k, \qquad U_c(\eta_{n,k} d_n) \le U_c(0) + c_1 \eta_{n,k} \nabla U_c \cdot d_n \tag{4}$$

Dimensional check: q, trust radius and U_c are dimensionless in the reach-normalized lane.

E5: Accepted-step integrity conditions

$$\widehat{H}_{n+1} \le \widehat{H}_n, \quad d_{\min,n+1} \ge f_c d_{\min,0}, \quad |\Delta Lk| \le \varepsilon_{Lk} \tag{5}$$

Dimensional check: Energy comparison is dimensionless; distance is compared as a ratio; linking is dimensionless.

E6: Cross-provider endpoint agreement

$$CV_H(K) = \frac{s(H_K^*)}{|\overline{H}_K^*|}, \qquad \Delta_R(K) = \frac{R^*_{\max} - R^*_{\min}}{|\overline{R}^*|} \tag{6}$$

Dimensional check: Coefficient of variation and relative spans are dimensionless.

E7: Constrained-objective Hessian with finite-difference sign convergence

$$K_H^{(j)} = \nabla_q^2 U_c(0; \epsilon/2^j), \quad j = 0, 1, 2; \qquad \text{promote sign} \iff s_0 = s_1 = s_2 \ \wedge \ \frac{\|K_H^{(2)} - K_H^{(1)}\|_F}{\max(\|K_H^{(2)}\|_F, \epsilon)} \leq \tau_H \tag{7}$$

Dimensional check: The reduced coordinates and U_c are dimensionless, so all Hessian entries/eigenvalues are dimensionless.

E8: Nonlinear reduced ringdown on the same constrained objective

$$\ddot{z} = -\partial_z U_c(z\, e_{\min}), \qquad e_{\min} = \underset{\|e\|=1}{\operatorname{argmin}}\, e^T K_H e \tag{8}$$

Dimensional check: Normalized time, z and U_c make the reduced test dimensionless.

E9: State-selected handoff predicate

$$Q_K = I[\text{provider consensus}] I[\sigma \leq \sigma_{\max}] I[K_H \succeq 0] I[\text{ringdown bounded}] \tag{9}$$

Dimensional check: Boolean predicate.

Registered code-to-mathematics steps

1. **S1** [G0; equations E1]: Verify frozen protocol/implementation and exact E013/E011 source-byte binding.
2. **S2** [G1; equations E1]: Reload only E013-frozen carriers and recheck geometry hashes.
3. **S3** [G2; equations E1, E3, E4, E5]: Validate rigid-energy invariance, exact constrained-objective evaluation, Armijo monotonicity and topology-proxy mechanics.
4. **S4** [G3; equations E2, E3, E4, E5]: Run the fixed constrained-objective variational selector independently on every current carrier; classify failure to demonstrate numerical descent as UNRESOLVED_NUMERICAL.
5. **S5** [G4; equations E3, E4]: Evaluate fixed-selector transfer on registered confirmation topologies.
6. **S6** [G5; equations E1]: Certify the common energy kernel against C++/OpenMP FP64.
7. **S7** [G6; equations E1]: Record optional GPU/DD32 framework screening without promotion.
8. **S8** [G7; equations E6]: Compare independently sourced endpoints topology-by-topology.
9. **S9** [G8; equations E7, E8]: Qualify stationary endpoints only after constrained-Hessian eps/2/4 convergence and then nonlinear ringdown on the same U_c.
10. **S10** [G10; equations E6]: Test whether initially provider-sensitive energy kernels collapse after selection.
11. **S11** [G11; equations E5]: Audit clearance/linking topology proxies over all paths.
12. **S12** [G12; equations E9]: Freeze only provider-converged stable endpoints as a future A057 handoff.

13. **S13** [G22; equations E3, E6, E7, E9]: Assign the blind overall state-selection classification under diagnostic continuation.

14. **S14** [G9; equations E9]: Reserve framework reveal integrity gate; v0.5.1 contains no mass target reveal.

7 Algorithm

Every E013 carrier is reloaded from exact source bytes and evaluated with one constrained objective $U_c(q) = \widehat{H}_a(\mathcal{N}[\mathcal{R}(X + \delta X(q))])$. The same U_c supplies finite-difference gradients, Armijo/trust-region acceptance, stationarity, endpoint Hessians, and nonlinear reduced ringdown. Accepted steps must preserve a nonlocal-clearance floor and pairwise linking within tolerance. Failure to demonstrate numerical descent is recorded as `UNRESOLVED_NUMERICAL`, not as negative physical evidence.

8 Sources

E011-v0.3.0 is the static-seed/provider authority; E013-v0.1.1 freezes the common carrier population; E010-v0.3.1 only reloads the exact source-native geometry bytes. No old A054 output is a science input.

9 Gates and diagnostic continuation

Only G0 is a hard prerequisite. Scientific failures do not suppress later diagnostics. Soft dependency caveats prevent a numerical diagnostic from being promoted beyond its support.

10 Numerics

BASIC, FULL and CERTIFY increase reduced basis richness, resolution, iteration budget and stationarity strictness. The same global Armijo/trust-region policy is used for every carrier; no provider/topology-specific optimization hyperparameter is fitted. Endpoint Hessians are checked at ϵ, $\epsilon/2$ and $\epsilon/4$ and their sign is promoted only after finite-difference convergence.

11 Backends

Python/NumPy FP64 is REFERENCE. C++/pybind11 OpenMP FP64 certifies the common line-energy kernel at relative-L2 $\leq 10^{-10}$. Framework SYCL/DD32 remains SCREENING_ONLY.

12 Blindness and provenance

There are no protected mass targets in v0.5.1. Blindness is used to lock the protocol and to exclude legacy scientific outputs. E013 carrier identities and provider groups remain visible because provider convergence is the object of the test.

13 Statistics

Primary statistics are stationarity success lower/upper bounds under explicit numerical-unresolved accounting, provider endpoint energy CV, ropelength relative span, absolute-writhe span, source-to-endpoint provider-CV collapse, Hessian finite-difference convergence/sign and nonlinear ringdown amplitude/energy drift.

14 Results

Automatic run provenance

Field	Recorded value
Protocol bundle SHA-256	a478e08e2bf47ee1703fa67e5a5da118aa1727e9ad627ba468c00bcbdd2feef1
Framework schema	SST-FALSIFIER-PROTOCOL-2
Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Output manifest SHA-256	97e03c4531c16345e64f59fbe167eb2c820eac25871b00b1c024ca2c4643323b

Gate ledger

Gate	Status	Question
G0	PASS	Is the frozen v0.5.1 protocol intact and runtime/implementation provenance self-cons
G1	PASS	Is the run exactly bound to the E013 common-carrier manifest, with all E011 source
G2	PASS	Do rigid-objectivity, constrained-objective identity, Armijo monotonicity and topolog
G3	UNRESOLVED	Can the fixed target-free constrained-objective selector drive the preregistered carrier
G4	UNRESOLVED	Does the same frozen constrained-objective selector generalize without refit to registe
G5	PASS	Does C++/OpenMP FP64 reproduce Python/NumPy FP64 for the state-selection e
G6	PASS	If available, does framework GPU/DD32 screening evidence remain consistent withou
G7	UNRESOLVED	Do independently sourced provider anchors of the same topology converge to the san
G8	UNRESOLVED	Are stationarity-qualified endpoints locally stable under a finite-difference-converged
G10	UNRESOLVED	For source-energy-sensitive topologies, does numerically resolved state selection colla
G11	PASS	Are clearance and link-drift topology proxies preserved along every completed state-s
G12	UNRESOLVED	Can at least one provider-converged stationary, Hessian-converged, stable state be fr
G22	UNRESOLVED	What is the overall state-selection evidence classification when numerical unresolved
G9	DEFERRED	After a completed FULL/CERTIFY blind run, do reveal and commitment checks ve

Backend manifest


```
{
  "cpp_certification": {
    "authority": "CERTIFICATION",
    "gate_status": "PASS"
  },
  "diagnostic_continuation": true,
  "gpu_screening": {
    "authority": "SCREENING_ONLY",
    "gate_status": "PASS"
  },
  "python_reference": {
    "actual_backend": "python-numpy-fp64",
    "authority": "REFERENCE",
    "precision": "float64"
  },
  "schema": "SST-BACKEND-MANIFEST-2",
  "scientific_gate_dependencies": "soft; see DEPENDENCY_CAVEATS.json"
}
```

Source manifest


```
{
  "manifest_sha256": "44be8e3e1053c49ef294c29e9946bab0e8aec78241e0bd17c936446578afee49",
  "schema": "SST-SOURCE-MANIFEST-2",
  "sources": [
    {
      "catalog_id": "E011",
      "evidence_class": "independent_source",
      "exists": true,
      "independence_family": "E011_SKLSA_v0.3.0",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\E_pipelines\\E011_sklsa_s
      "provenance_family": "E011_SKLSA_STATIC_READY",
      "registry_hint": {
        "catalog_id": "E011",
        "hits": []
      },
      "role": "static_seed_qualification_authority",
      "sha256": "eda7cdcddaf851d6e83b225df5182634a70b9678bfe3d3a0f338a28c2a0e8668",
      "source_id": "S_E011_STATIC_INDEX",
      "version": "v0.3.0"
    },
    {
      "catalog_id": "E011",
      "evidence_class": "independent_source",
      "exists": true,
      "independence_family": "E011_SKLSA_v0.3.0",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\E_pipelines\\E011_sklsa_s
      "provenance_family": "E011_SKLSA_STATIC_READY",
      "registry_hint": {
        "catalog_id": "E011",
        "hits": []
      },
      "role": "provider_anchor_authority",
      "sha256": "999739dad02aa2b4dbc6c00411c9e21cfc59252de5daff65eb6471a168782d7d",
      "source_id": "S_E011_PROVIDER_ANCHORS",
      "version": "v0.3.0"
    },
    {
      "catalog_id": "E013",
      "evidence_class": "simulation",
      "exists": true,
      "independence_family": "E013_protocol",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\E_pipelines\\E013_PKLSA_C
      "provenance_family": "E013_PKLSA_Cross_Falsifier",
      "registry_hint": {
        "catalog_id": "E013",
        "hits": []
      },
      "role": "cross_falsifier_common_carrier_protocol",
      "sha256": "162a65d09d431ebee4943a1f348770f6f7c442d257815f96d6bcb6cf2377f187",
      "source_id": "S_E013_CAMPAIGN_CONFIG",
      "version": "v0.1.1"
```

```
    },
    {
      "catalog_id": "E010",
      "evidence_class": "simulation",
      "exists": true,
      "independence_family": "E010_implementation",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\E_pipelines\\E010_pklsa_p
      "provenance_family": "E010_PKLSA_v0.3.1",
      "registry_hint": {
        "catalog_id": "E010",
        "hits": []
      },
      "role": "source_native_geometry_parser",
      "sha256": "597d5e3b06b0a91884edd2a7ffa99b2c7160940c68c9da39addb0d0c32796d26",
      "source_id": "S_E010_LOADER",
      "version": "v0.3.1"
    }
  ],
  "workbench_root": "C:\\workspace\\projects\\SST-Workbench"
}
```


15 Interpretation

A stationary point is not a stable state merely because $\nabla U_c \simeq 0$. Provider convergence and positive local stability are independent requirements. Numerical inability to establish descent or Hessian-sign convergence remains unresolved. Only topologies satisfying provider convergence, stationarity, converged nonnegative local stability and bounded ringdown are eligible for the state-selected handoff.

16 Reproducibility

Run `run_00_install.cmd`, then execute through E013 with `run_all_cross_falsifier.cmd BASIC/FULL/CERTIF`. The first A054 run automatically freezes the create-once protocol.

References

[1] H. K. Moffatt, "The degree of knottedness of tangled vortex lines," *J. Fluid Mech.* 35, 117–129 (1969). DOI: 10.1017/S0022112069000991.

[2] P. G. Saffman, *Vortex Dynamics*. Cambridge University Press (1992). DOI: 10.1017/CBO9780511624063.

[3] R. L. Bishop, "There is More than One Way to Frame a Curve," *Amer. Math. Monthly* 82, 246–251 (1975). DOI: 10.1080/00029890.1975.11993807.